
GENERAL RELATIVITY

Tutorial problem set 17, 29.01.2021, with solutions.

■ PROBLEM 41 Response of matter to gravitational waves.

The equation of motion for the physical degrees of freedom of gravity propagating in Minkowski space-time is

$$\square h_{ij}^{TT} = 0 , \quad (41.1)$$

where h_{ij}^{TT} is a transverse traceless tensor,

$$\partial^i h_{ij}^{TT} = 0 , \quad (h^{TT})^i_i = 0 . \quad (41.2)$$

Equation of motion (41.1) obviously has propagating waves as solutions.

(a) Assume a plane wave propagating in an arbitrary direction,

$$h_{ij}^{TT} = A_\sigma \epsilon_{ij}^\sigma \cos \left(\omega t - \vec{k} \cdot \vec{x} \right) . \quad (41.3)$$

where $\sigma = +, \times$ are the two polarization tensors. Determine the dispersion relation of the wave in (41.3).

Now assume a wave propagating in the positive z -direction ($\vec{k} = k\hat{z}$) and show that there are two independent polarizations of a gravitational wave, a *plus* one and a *cross* one, corresponding to polarization tensors

$$\epsilon^+ = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} , \quad \epsilon^\times = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} . \quad (41.4)$$

Next we want to determine what is the effect this wave has on a group of test particles initially distributed in a circle perpendicular to the propagation direction of the wave. In order to do that we will solve the geodesic deviation equation. The geodesic deviation equation for a separation vector S^μ between two nearby particle trajectories reads,

$$\frac{D^2}{d\tau^2} S^\mu = R^\mu{}_{\nu\rho\sigma} \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} S^\sigma . \quad (41.5)$$

(b) Assume the test particles to move slowly, $dx^\alpha/d\tau \approx (1, 0, 0, 0)$. Show that by expanding the left- and the right-hand side of equation (41.5) to the lowest non-vanishing order the geodesic deviation equation becomes,

$$\frac{\partial^2}{\partial t^2} S_i - \frac{1}{2} (\partial_t^2 h_{ij}^{TT}) S_j = \frac{1}{2} S^j \frac{\partial^2}{\partial t^2} h_{ij}^{TT} . \quad (41.6)$$

Show that – to the linear order in h_{ij}^{TT} , this equations can be rewritten as,

$$\frac{\partial^2}{\partial t^2} \left[S_i - \frac{1}{2} h_{ij}^{TT} S_j \right] = \frac{1}{2} \frac{\partial^2}{\partial t^2} h_{ij}^{TT} \left[S_j - \frac{1}{2} h_{jl}^{TT} S_l \right] , \quad (41.7)$$

and that $\bar{S}_i \equiv S_i - \frac{1}{2} h_{ij}^{TT} S_j$ can be interpreted as the invariant distance that detectors measure.

- (c) Consider a gravitational wave as in (41.3) with $A_{\times} = 0$. Show that (41.6) has the following perturbative solution for a wave propagating in the z -direction,

$$\bar{S}^1 = \left(1 + \frac{A_+}{2} \cos(\omega t - kz)\right) \bar{S}^1(0) , \quad (41.8)$$

$$\bar{S}^2 = \left(1 - \frac{A_+}{2} \cos(\omega t - kz)\right) \bar{S}^2(0) , \quad (41.9)$$

where A_+ serves as a perturbation parameter in equation (41.6). Sketch how the particles move assuming they were distributed in a circle at the initial moment.

- (d) Repeat the analysis in (c) for $A_+ = 0$. $A_{\times} \neq 0$.

SOLUTION: (a) Inserting (41.3) into (41.1) gives

$$\left(\omega^2 - \|\vec{k}\|^2\right) A_{\sigma} \epsilon_{ij}^{\sigma} \cos\left(\omega t - \vec{k} \cdot \vec{x}\right) = 0 , \quad (41.10)$$

from which one obtains the dispersion relation for gravitational waves,

$$\omega = \pm \|\vec{k}\| , \quad (41.11)$$

where the plus sign (minus sign) refers to the wave with phase velocity in the direction of motion (opposite of the direction of motion) of the wave.

To address the second part of the question, note that gravitational waves must satisfy the two gauge conditions (41.2), which fix the polarization tensors (41.4). For a wave moving in the z direction, we have $\vec{k} \cdot \vec{x} = kz$ and Eq. (41.3) simplifies to,

$$h_{ij}^{TT}(t, z) = A_+ \epsilon_{ij}^+ \cos(\omega t - kz) + A_{\times} \epsilon_{ij}^{\times} \cos(\omega t - kz) . \quad (41.12)$$

Note first that $h_{ij}^{TT}(t, z)$ is traceless, $\delta_{ij} h_{ij}^{TT}(t, z) = 0$ thanks to the traceless nature of ϵ^+ and $\epsilon^{\times}$ in (41.4), $\delta_{ij} \epsilon_{ij}^+ = 0$ and $\delta_{ij} \epsilon_{ij}^{\times} = 0$. Next, acting ∂_i on $h_{ij}^{TT}(t, z)$ gives,

$$\begin{aligned} \partial_i h_{ij}^{TT}(t, z) &= A_+ \epsilon_{ij}^+ k \delta_{iz} \sin(\omega t - kz) + A_{\times} \epsilon_{ij}^{\times} k \delta_{iz} \sin(\omega t - kz) \\ &= k A_+ \epsilon_{zj}^+ \sin(\omega t - kz) + k A_{\times} \epsilon_{zj}^{\times} \sin(\omega t - kz) = 0 . \end{aligned} \quad (41.13)$$

- (b) Inserting $dx^{\alpha}/d\tau \approx (1, 0, 0, 0)$ into (41.5) and upon lowering index μ yields,

$$\frac{D^2 S_{\mu}}{d\tau^2} = \frac{d}{d\tau} \frac{D S_{\mu}}{d\tau} - \Gamma_{\mu\nu}^{\alpha} \frac{dx^{\nu}}{d\tau} \frac{D S_{\alpha}}{d\tau} = \frac{d^2 S_{\mu}}{d\tau^2} - (\partial_{\beta} \Gamma_{\alpha\mu}^{\rho}) \frac{dx^{\alpha}}{d\tau} \frac{dx^{\beta}}{d\tau} S_{\rho} + \mathcal{O}(h_{\mu\nu}^2) = R_{\mu 00\sigma} S^{\sigma} . \quad (41.14)$$

To get to the second equality, we have dropped the second order terms in the connection Γ and the terms of the form, $\Gamma \partial S$, as ∂S and Γ are both of the order of h , and thus of the first order, and we made use of the geodesic equation to replace $\ddot{x}$ by $-\Gamma \dot{x}^2$. Next, to the leading (linear) order in gravitational waves $h_{\mu\nu}$, the Levi-Civita connection reads,

$$\Gamma_{\alpha\beta}^{\rho} = \partial_{(\alpha} h_{\beta)}^{\rho} - \frac{1}{2} \partial^{\rho} h_{\alpha\beta} , \quad \partial^{\rho} = \eta^{\rho\sigma} \partial_{\sigma} , \quad h_{\beta}^{\rho} = \eta^{\rho\sigma} h_{\sigma\beta} . \quad (41.15)$$

from where we see that the nonvanishing components in the TT gauge are,

$$\Gamma_{ij}^0 = \frac{1}{2} \partial_0 h_{ij}^{TT} = \Gamma_{0j}^i = \Gamma_{j0}^i , \quad \Gamma_{jl}^i = \frac{1}{2} [\partial_j h_{il}^{TT} + \partial_l h_{ij}^{TT} - \partial_i h_{jl}^{TT}] . \quad (41.16)$$

Inserting this into (41.14) one obtains,

$$\frac{d^2 S_0}{d\tau^2} - (\partial_0 \Gamma_{00}^\rho) S_\rho = \frac{d^2 S_0}{d\tau^2} = 0 \quad (41.17)$$

$$\frac{d^2 S_i}{d\tau^2} - (\partial_0 \Gamma_{0i}^\rho) S_\rho = \frac{d^2 S_i}{d\tau^2} - (\partial_0 \Gamma_{0i}^j) S_j = R_{i00j} S^j. \quad (41.18)$$

Therefore we see that – to the leading order – gravitational waves do not affect S_0 , which for convenience can be set to zero.

From problem 40 in set 16 we know the Riemann tensor,

$$R_{\mu\nu\rho\sigma} = \frac{1}{2} \left(\partial_\rho \partial_\nu h_{\mu\sigma} + \partial_\sigma \partial_\mu h_{\nu\rho} - \partial_\rho \partial_\mu h_{\nu\sigma} - \partial_\sigma \partial_\nu h_{\mu\rho} \right). \quad (41.19)$$

We need R_{i00j} ,

$$R_{i00j} = \frac{1}{2} \left(\partial_0^2 h_{ij} + \partial_i \partial_j h_{00} - \partial_0 \partial_i h_{0j} - \partial_j \partial_0 h_{i0} \right) = \frac{1}{2} \partial_0^2 h_{ij}^{TT}. \quad (41.20)$$

With these we can rewrite the equation for S_i as,

$$\frac{d^2 S_i}{d\tau^2} - \frac{1}{2} (\partial_t^2 h_{ij}^{TT}) S_j = \frac{1}{2} (\partial_t^2 h_{ij}^{TT}) S^j. \quad (41.21)$$

In order to understand the significance of the term $-\frac{1}{2} (\partial_t^2 h_{ij}^{TT}) S_j$ on the left hand side, notice that one can use integration by parts twice to rewrite it as,

$$\begin{aligned} -\frac{1}{2} (\partial_t^2 h_{ij}^{TT}) S_j &= -\frac{1}{2} \partial_t [(\partial_t h_{ij}^{TT}) S_j] + \frac{1}{2} (\partial_t h_{ij}^{TT}) \partial_t S_j = -\frac{1}{2} \partial_t [(\partial_t h_{ij}^{TT}) S_j] + \mathcal{O}(h_{ij}^2) \\ &= -\frac{1}{2} \partial_t^2 (h_{ij}^{TT} S_j) + \mathcal{O}(h_{ij}^2). \end{aligned} \quad (41.22)$$

Upon inserting this into (41.21), it becomes,

$$\frac{d^2}{d\tau^2} \left[S_i - \frac{1}{2} h_{ij}^{TT} S_j \right] = \frac{1}{2} (\partial_t^2 h_{ij}^{TT}) S^j. \quad (41.23)$$

But, for small distances $\bar{S}_i \equiv S_i - \frac{1}{2} h_{ij}^{TT} S_j$ is precisely the invariant distance one measures. To see that, note that for $S_0 = 0$, $S^2 = g^{\mu\nu} S_\mu S_\nu = S_i^2 - h_{ij}^{TT} S_i S_j + \mathcal{O}(h_{ij}^2) \equiv \bar{S}_i^2$ is the invariant distance squared. Taking a square root of this expression, we see that the invariant distance in the i th direction is given by,

$$\bar{S}_i = S_i - \frac{1}{2} h_{ij}^{TT} S_j. \quad (41.24)$$

Indeed, squaring this yields,

$$\bar{S}_i \bar{S}_i = \left(S_i - \frac{1}{2} h_{ij}^{TT} S_j \right) \left(S_i - \frac{1}{2} h_{il}^{TT} S_l \right) = S_i^2 - h_{ij}^{TT} S_i S_j + \mathcal{O}(h_{ij}^2) \quad (41.25)$$

With these results in mind, one can recast Eq. (41.21) in the form,

$$\frac{d^2 \bar{S}_i}{d\tau^2} = \frac{1}{2} (\partial_t^2 h_{ij}^{TT}) \bar{S}^j + \mathcal{O}(h_{ij}^2). \quad (41.26)$$

(c) Here we are interested in a gravitational wave of the + polarization, $h_{ij}^{TT}(t, z) = A_+ \epsilon_{ij}^+ \cos(\omega t - kz)$, for which the Riemann tensor, $R_{i00j} = \frac{1}{2} \partial_t^2 h_{ij}^{TT}(t, z) = -\frac{\omega^2}{2} h_{ij}^{TT}(t, z)$. Upon inserting this into (41.21) one obtains (recall that $\bar{S}_i = \bar{S}^i$),

$$\partial_t^2 \bar{S}^i = -\frac{\omega^2}{2} A_+ \epsilon_{ij}^+ \cos(\omega t - kz) \bar{S}^j, \quad (41.27)$$

or in components,

$$\begin{aligned} \partial_t^2 \bar{S}^1 &= -\frac{\omega^2}{2} A_+ \cos(\omega t - kz) \bar{S}^1 \\ \partial_t^2 \bar{S}^2 &= \frac{\omega^2}{2} A_+ \cos(\omega t - kz) \bar{S}^2 \\ \partial_t^2 \bar{S}^3 &= 0, \end{aligned} \quad (41.28)$$

from where we see that $\bar{S}^3(t) = \bar{S}^3(0)$. To solve for $\bar{S}^1$ and $\bar{S}^2$, note that we can safely assume that $\bar{S}^i(0) \gg A_+$ and hence we can find a perturbative solution in powers of A_+ ,

$$\bar{S}^i = \bar{S}_{(0)}^i + \bar{S}_{(1)}^i + \cdots, \quad (41.29)$$

where $\bar{S}_{(0)}^i = \mathcal{O}(A_+^0)$ and $\bar{S}_{(1)}^i = \mathcal{O}(A_+)$. Inserting the *Ansatz* (41.29) into (41.28) one obtains,

$$\begin{aligned} \partial_t^2 \bar{S}_{(1)}^1 &= -\frac{\omega^2}{2} A_+ \cos(\omega t - kz) \bar{S}_{(0)}^1 \implies \bar{S}_{(1)}^1 = \frac{A_+}{2} \cos(\omega t - kz) \bar{S}_{(0)}^1 \\ \partial_t^2 \bar{S}_{(1)}^2 &= \frac{\omega^2}{2} A_+ \cos(\omega t - kz) \bar{S}_{(0)}^2 \implies \bar{S}_{(1)}^2 = -\frac{A_+}{2} \cos(\omega t - kz) \bar{S}_{(0)}^2, \end{aligned} \quad (41.30)$$

which agrees with Eq. (41.9). The circle of particle oscillates into an ellipse which is at $t = 0$ elongated along the x axis by $+A_+ \bar{S}^1(0)/2$ and compressed along the y axis by $-A_+ \bar{S}^2(0)/2$. After a quarter period, at $t = \pi/(2\omega)$, one observes a circle, at $t = \pi/\omega$, one observes an ellipse which is compressed along the x axis by $-A_+ \bar{S}^1(0)/2$ and elongated along the y axis by $+A_+ \bar{S}^2(0)/2$, *etc.*

(d) In this case we ought to consider gravitational waves of the $\times$ polarization, $h_{ij}^{TT}(t, z) = A_\times \epsilon_{ij}^\times \cos(\omega t - kz)$, and the corresponding equation of motion for $\bar{S}^i$ is,

$$\partial_t^2 \bar{S}^i = -\frac{\omega^2}{2} A_\times \epsilon_{ij}^\times \cos(\omega t - kz) \bar{S}^j, \quad (41.31)$$

or in components,

$$\begin{aligned} \partial_t^2 \bar{S}^1 &= -\frac{\omega^2}{2} A_\times \cos(\omega t - kz) \bar{S}^2 \\ \partial_t^2 \bar{S}^2 &= \frac{\omega^2}{2} A_\times \cos(\omega t - kz) \bar{S}^1 \\ \partial_t^2 \bar{S}^3 &= 0, \end{aligned} \quad (41.32)$$

from where we see that $\bar{S}^3(t) = \bar{S}^3(0)$. We make the same *Ansatz* as above,

$$\bar{S}^i = \bar{S}_{(0)}^i + \bar{S}_{(1)}^i + \cdots, \quad (41.33)$$

where $\bar{S}_{(0)}^i = \mathcal{O}(A_\times^0)$ and $\bar{S}_{(1)}^i = \mathcal{O}(A_\times)$. Inserting the *Ansatz* (41.33) into (41.32) one obtains,

$$\begin{aligned}\partial_t^2 \bar{S}_{(1)}^1 &= -\frac{\omega^2}{2} A_\times \cos(\omega t - kz) \bar{S}_{(0)}^2 \implies \bar{S}_{(1)}^1 = \frac{A_\times}{2} \cos(\omega t - kz) \bar{S}_{(0)}^2 \\ \partial_t^2 \bar{S}_{(1)}^2 &= -\frac{\omega^2}{2} A_\times \cos(\omega t - kz) \bar{S}_{(0)}^1 \implies \bar{S}_{(1)}^2 = \frac{A_\times}{2} \cos(\omega t - kz) \bar{S}_{(0)}^1.\end{aligned}\quad (41.34)$$

This solution deforms a circle of particles as for the + polarization, but now rotated in the xy plane by 45° .

■ **PROBLEM 42** Gravitational radiation from a binary star system.

Two stars of masses m_1 and m_2 are orbiting each other in non-relativistic circular orbit, separated by a distance R . Since this system loses energy due to gravitational radiation, the separation distance will change in time, $R = R(t)$.

(a) Show that the total (Newtonian) energy of the system is

$$E = -\frac{1}{2} \frac{G_N M \mu}{R}, \quad (42.1)$$

where $M = m_1 + m_2$ and $\mu = \frac{m_1 m_2}{m_1 + m_2}$. Assume that the orbit is a circular one and the separation between the stars is $R = \text{const.}$

(b) Show that the power P_{GW} dissipated by gravitational waves is

$$P_{GW} = \frac{32}{5} \frac{G_N^4 \mu^2 M^3}{R^5}, \quad (42.2)$$

Use the following formula derived in the lectures

$$P_{GW} = \frac{G_N}{5} \langle \partial_0^3 t_{ij} \partial_0^3 t^{ij} \rangle, \quad (42.3)$$

where

$$t_{ij} = \int d^3x T_{00} \left(x_i x_j - \frac{\delta_{ij}}{3} \vec{x}^2 \right). \quad (42.4)$$

(c) The gravitational power is radiated at the expense of orbital energy. Show that the separation between the stars depends on time as

$$R(t) = \left[\frac{256}{5} M^2 \mu (t_0 - t) \right]^{1/4}, \quad (42.5)$$

if we assume the energy loss to be slow enough that the approximation of instantaneously circular orbits holds throughout. Here t_0 is the time when the separation distance goes to zero. Find this time in terms of the masses, and the initial separation distance R_0

In 1974 Hulse and Taylor discovered a binary system PSR1913+16, in which both stars are very small, and one of them is a pulsar. The period of the orbit was about eight hours, extremely small by astrophysical standards, and the fact that one of the stars was a pulsar provided a very accurate clock with respect to which the change in the period of the orbit as

the system loses energy can be measured. The result was consistent with the prediction from General Relativity for the energy loss through gravitational radiation. Hulse and Taylor were awarded the Nobel prize in 1993 for their work.

SOLUTION: (a) Let us begin by observing that the centrifugal forces acting on m_1 and m_2 are balanced by the gravitational force,

$$m_1 r_1 \Omega^2 = m_2 r_2 \Omega^2 = \frac{G_N m_1 m_2}{R^2}, \quad (42.6)$$

where $\Omega = v_1/r_1 = v_2/r_2$ is the angular velocity and $R = r_1 + r_2$ is the distance between m_1 and m_2 and r_1 and r_2 are the corresponding distances from the center of mass. Eq. (42.6) implies,

$$r_2 = \frac{m_1}{m_2} r_1, \quad r_1 = \frac{m_2}{m_1 + m_2} R, \quad r_2 = \frac{m_1}{m_1 + m_2} R, \quad \Omega^2 = \frac{G_N (m_1 + m_2)}{R^3}. \quad (42.7)$$

One can use these results in the energy of the system to obtain,

$$\begin{aligned} E &= \frac{m_1 r_1^2 \Omega^2}{2} + \frac{m_2 r_2^2 \Omega^2}{2} - \frac{G_N m_1 m_2}{R} = \frac{G_N m_1 m_2^2}{2(m_1 + m_2)R} + \frac{G_N m_1^2 m_2}{2(m_1 + m_2)R} - \frac{G_N m_1 m_2}{R} \\ &= -\frac{G_N m_1 m_2}{2R} = -\frac{G_N \mu M}{2R}. \end{aligned} \quad (42.8)$$

(b) The (00) component of the energy momentum tensor is,

$$T_{00} = m_1 \delta(x - r_1 \cos(\Omega t)) \delta(y - r_1 \sin(\Omega t)) \delta(z) + m_2 \delta(x + r_2 \cos(\Omega t)) \delta(y + r_2 \sin(\Omega t)) \delta(z), \quad (42.9)$$

where we assumed that the speeds $v_1 = r_1 \Omega$, $v_2 = r_2 \Omega$ are nonrelativistic. Inserting this into (42.4) gives,

$$\begin{aligned} t_{ij} &= \int dx dy \left[m_1 \delta(x - r_1 \cos(\Omega t)) \delta(y - r_1 \sin(\Omega t)) \right. \\ &\quad \left. + m_2 \delta(x + r_2 \cos(\Omega t)) \delta(y + r_2 \sin(\Omega t)) \right] \left(x_i x_j - \frac{\delta_{ij}}{3} \vec{x}^2 \right). \end{aligned} \quad (42.10)$$

After performing the integrals one gets for the different components of the moment of inertia,

$$\begin{aligned} t_{xx} &= m_1 \left[r_1^2 \cos^2(\Omega t) - \frac{1}{3} r_1^2 \right] + m_2 \left[r_2^2 \cos^2(\Omega t) - \frac{1}{3} r_2^2 \right] = \frac{\mu R^2}{2} \left[\cos(2\Omega t) + \frac{1}{3} \right] \\ t_{yy} &= m_1 \left[r_1^2 \sin^2(\Omega t) - \frac{1}{3} r_1^2 \right] + m_2 \left[r_2^2 \sin^2(\Omega t) - \frac{1}{3} r_2^2 \right] = \frac{\mu R^2}{2} \left[-\cos(2\Omega t) + \frac{1}{3} \right] \\ t_{zz} &= m_1 \left[-\frac{1}{3} r_1^2 \right] + m_2 \left[-\frac{1}{3} r_2^2 \right] = \mu R^2 \left[-\frac{1}{3} \right] \\ t_{xy} &= t_{yx} = m_1 \left[r_1^2 \cos(\Omega t) \sin(\Omega t) \right] + m_2 \left[r_2^2 \cos(\Omega t) \sin(\Omega t) \right] = \frac{\mu R^2}{2} [\sin(2\Omega t)] \\ t_{xz} &= t_{zx} = 0 \\ t_{yz} &= t_{zy} = 0. \end{aligned} \quad (42.11)$$

and their third time derivatives are,

$$\begin{aligned}
\partial_t^3 t_{xx} &= 4\Omega^3 \mu R^2 \sin(2\Omega t) \\
\partial_t^3 t_{yy} &= -4\Omega^3 \mu R^2 \sin(2\Omega t) \\
\partial_t^3 t_{zz} &= 0 \\
\partial_t^3 t_{xy} &= -4\Omega^3 \mu R^2 \cos(2\Omega t) \\
\partial_t^3 t_{xz} &= \partial_t^3 t_{zx} = 0 \\
\partial_t^3 t_{yz} &= \partial_t^3 t_{zy} = 0 .
\end{aligned} \tag{42.12}$$

Inserting these into Eq. (42.3) yields

$$P_{GW} = \frac{16G_N\Omega^6\mu^2R^4}{5} \langle 2 \rangle = \frac{32}{5} \frac{G_N^4\mu^2M^3}{R^5} = -\frac{dE}{dt} , \tag{42.13}$$

which agrees with (42.3).

(c) Energy must be conserved, hence the energy emitted by gravitational waves (42.13) is on the expense of the orbital energy (42.8),

$$\frac{G_N\mu M}{2R^2} \frac{dR}{dt} + \frac{32}{5} \frac{G_N^4\mu^2M^3}{R^5} = 0 \implies \frac{dR^4}{dt} + \frac{256}{5} G_N^3\mu M^2 = 0 , \tag{42.14}$$

whose solution is,

$$R(t) = \left[\frac{256}{5c^5} G_N^3 M^2 \mu (t_0 - t) \right]^{1/4} , \tag{42.15}$$

where $t = 0$ is the initial time at which, $R(t = 0) = R_0 = \left[\frac{256}{5c^5} G_N^3 M^2 \mu t_0 \right]^{1/4}$ and t_0 is the merger time at which $R(t_0) = 0$, or

$$t_0 = \frac{5R_0^4 c^5}{256G_N^3 M^2 \mu} , \tag{42.16}$$

where c is the speed of light.